neutron velocity. The average number of collisions n required to reduce the neutron energy from E_n^0 to E_n is given by

$$n = \xi^{-1} \ln\left(E_n^0/E_n\right) + 1 \tag{19.8}$$

The *slowing down power* (*SDP*) of a moderator depends in addition on the neutron scattering cross section and number of scattering atoms per unit volume (N_0):

$$SDP = \xi\, N_0\, \sigma_s = \xi\, \Sigma_s \tag{19.9}$$

The *SDP* is an average value over the epithermal neutron energy region. A good neutron moderator should divert few neutrons from the fission process, i.e. the neutron absorption cross section must be small. In this respect both heavy water (i.e. D_2O) and carbon are superior to H_2O, since for H_2O the reaction probability for $^1H(n,\gamma)^2H$ is relatively large (Table 19.3). In order to incorporate this property, the concept *moderating ratio* (*MR*) is used as a criterion of the moderator property in the thermal neutron energy region:

$$MR = SDP/\Sigma_a = \xi\, \Sigma_s/\Sigma_a \tag{19.10}$$

The moderating properties are summarized in Table 19.3. The moderator qualities decrease in the order $D_2O > C > Be > H_2O$. In a commercial reactor, price and other properties must also be considered, so H_2O is favoured over D_2O in most reactors.

19.6. THE NEUTRON CYCLE

In order for a chain reaction to occur at least one of the neutrons released in fission must produce a new fission event. This condition is defined by the *multiplication factor k*:

$$k = \frac{\text{number of neutrons in generation 2}}{\text{number of neutrons in generation 1}} \tag{19.11}$$

when $k > 1$ the number of neutrons in the second generation exceeds the number of neutrons consumed. Under this condition, the neutron flux, and, thus, the number of fission events increase for each successive neutron generation with a resultant increase in the power production of the reactor (§19.8 discusses further the *reactor kinetics*). If $k = 1$ the number of fissions per unit time, and thus the energy production, is constant. For $k < 1$ the chain reaction cannot be maintained. A reactor operating with $k = 1$ is said to be *critical*, $k > 1$ *supercritical*, and $k < 1$ *subcritical*. k is regulated by means of the control rods and in PWRs also by the n-absorber concentration in the water (usually boric acid).

In any neutron generation the neutrons experience a variety of fates (Fig. 19.6). Some neutrons escape the reactor and some are absorbed in the reactor structural materials and shielding (i.e. fuel cans, control rods, moderator, coolant, etc.). To take this loss into

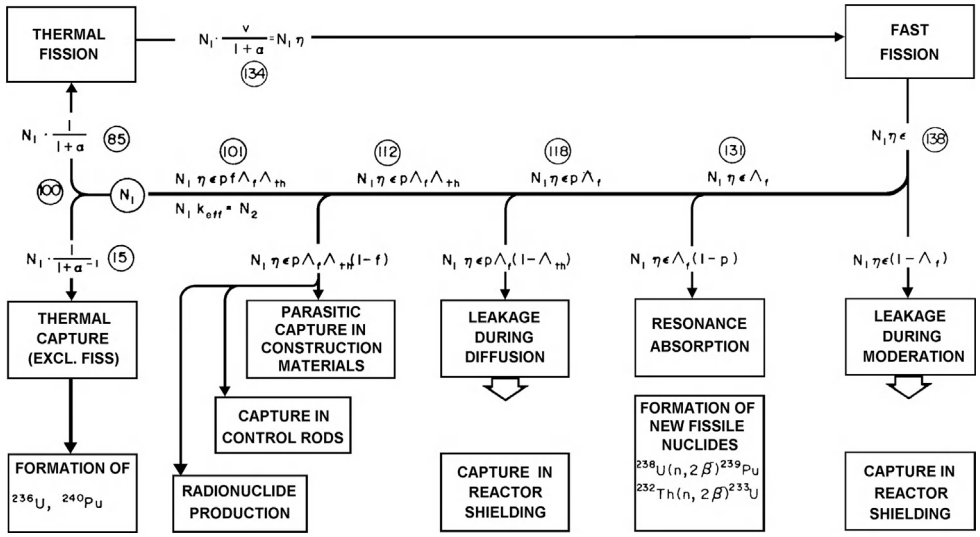


Figure 19.6 The neutron cycle in a thermal reactor. The number of neutrons for 100 in first generation are given in circles. *(data from an old Belgian research reactor, BR1)*

account, two different multiplication factors are used: k_∞ refers to a reactor of infinite dimensions (i.e. no leakage) while k_{eff} refers to a reactor of a finite size:

$$k_{\text{eff}} = k_\infty \Lambda \tag{19.12}$$

Λ is the fraction of neutrons which are *not* lost through leakage to the surroundings (*the non-leakage factor*). In order to minimize the neutron leakage, the reactor core is surrounded with a *neutron reflector* which for thermal neutrons in LWRs is water (graphite or beryllium are sometimes used in other reactor designs); for fast neutron reflection iron is frequently used.

Let us consider the neutron cycle in a reactor, i.e. the fate of the neutrons as they proceed from one generation to the next (Figure 19.6, where leakage, treated in §19.7, also is included). We start by assuming that we have an air cooled reactor containing natural uranium with a graphite moderator in a configuration which allows negligible leakage ($\Lambda = 1$). If 100 neutrons (N_1) are captured by the uranium fuel, the fission releases $N_1\,\eta$ or $100 \times 1.34 = 134$ new neutrons. The fast neutrons may also cause some fission in nearby uranium atoms, thereby releasing more neutrons. This is measured by ε, *the fast fission factor*. Its value in a heterogeneous nuclear reactor depends to a large extent on the moderator and on the dimensions within a fuel element. For natural uranium in graphite ε is about 1.03, so that the 134 neutrons released in fission by the original 100 thermal neutrons are increased to $N_1\,\eta\,\varepsilon$ (or $1.03 \times 134 = 138$) neutrons. In homogeneous reactors $\varepsilon = 1$ because the fission neutrons do not have to pass through any zone with concentrated fuel to reach the moderator.

These energetic neutrons are slowed down through collisions with surrounding atoms and decrease steadily in energy until thermal energies are reached. In the energy range 1 to 10^5 eV, *the epithermal region*, ^{235}U and ^{238}U have large resonance peaks for radiative capture, while fission in ^{235}U dominates at thermal energies (Fig. 19.3). In order to maximize the fission probability, the losses through radiative capture in the epithermal region must be minimized. This can be achieved by physically separating moderator and fuel. The fast neutrons, which easily escape from thin fuel pins, are mainly slowed in the surrounding moderator. Because the moderator is of a material with light atoms, the number of collisions required to slow these neutrons is small. Therefore, it is unlikely that the neutrons encounter a fuel pin (and are captured) until thermal energies have been reached. The probability that the neutrons pass through the energy region of the resonance peaks without capture is called the *resonance escape probability* and denoted by p. From the original N_1 neutrons at this point there would be $N_1\,\eta\,\varepsilon\,p$ neutrons. In a reactor of natural uranium and graphite, p is usually around 0.9. Therefore, from the original 100 neutrons 138×0.9 or 124 second generation neutrons reach the thermal energy range.

The cross sections for neutron capture increase for all atoms for thermal energy neutrons. As a result, even though low cross section materials are used some neutrons are captured by the structural and moderator materials. The probability for the non-capture of thermal neutrons in this fashion is signified by f, *the thermal utilization factor*, which in our case can be assumed to be $\sim$0.9. Thus of the original N_1 neutrons 112 thermal neutrons remain in the second generation to cause fission in the nuclear fuel.

These 112 thermal neutrons constitute the second generation of neutrons which according to our definition of neutron multiplication factor is k_∞ so that

$$k_\infty = \eta\,\varepsilon\,p\,f \tag{19.13}$$

This relation is known as the *four factor formula*. In our case $k_\infty = 1.12$.

The value of η depends on how much fissile material the fuel contains. The other three factors have a more complicated dependence of the reactor design including the fuel/moderator ratio, their amounts and shape; some values are given for a practical case in Table 19.4.

For simplicity, let us for a moment regard a homogeneous reactor. For such a reactor

$$f = \Sigma_\mathrm{a}\,(\mathrm{fuel})/[\Sigma_\mathrm{a}\,(\mathrm{fuel}) + \Sigma_\mathrm{a}\,(\mathrm{mod}) + \Sigma_\mathrm{a}\,(\mathrm{other})] \tag{19.14}$$

where the denominator is the macroscopic cross section of all absorption reactions in the reactor; the last term includes absorption by impurities in construction materials and by products formed in the course of the operation, e.g. fission products. When control rods are inserted Σ_a (other) increases. Effective cross sections, valid for the current neutron energy spectrum, must be used for all nuclides.

In order to allow for a decrease in f during operation, commercial reactors are designed with k_∞ 1.2−1.3, rather than the 1.12 given in the example above. Even

Table 19.4 Data for a typical BWR power plant

General: 640 MW_e (1912 MW_{th}) BWR plant at Würgassen, Germany, in commercial operation 1975–1995.

Reactor physics: θ 4.5 × 10^5 (s), η 1.740, f 0.865, p 0.811, ε 1.041, k_∞ 1.270, k_{eff} 1.255, L^2 3.67 (cm^2), τ 34.83 (cm^2), B^2 2.32 × 10^{-4} (cm^{-2}), $\phi_{th,average}$ 4.4 × 10^{13} (cm^{-2} s^{-1}), $\phi_{fast,average}$ 1.9 × 10^{14} (cm^{-2} s^{-1}), Δk_{max} 25.5%, $\Delta k/k$ 2.2 × 10^{-3}%/°C

Core: 86 600 kg U, 2.6% ^{235}U; critical mass 38.5 kg ^{235}U; burn-up 27 500 MWd t^{-1}; refueling 1/5 of core annually; storage pool for 125 t fuel; core volume 38 m^3.

Fuel element: UO_2 pellets, 1.43 cm diameter; 49 rods per assembly; 444 assemblies in core; enrichment 2.6% ^{235}U; Gd_2O_3 burnable poison; zircaloy cladding 0.8 mm.

Heat and coolant data: Max. fuel temp. 2580°C, max. can temp. 370°C; coolant inlet temp. 190°C, outlet 285°C, pressure 7 MPa (70 bar); 280 t coolant, 2 × 5200 m^3 h^{-1} internal pump capacity; condenser cooling water 95 000 t h^{-1} (River Weser).

higher k_∞ values occur as is the case for reactors with highly enriched fuel. In reactors with large k_∞, which permits high burnup values, a small amount of suitable n-absorbers (*poisons*) may be included in the fuel at the start to decrease k_∞. Such a poison is gadolinium, especially ^{157}Gd, which occurs to 16% in natural gadolinium and has a thermal neutron cross section of 254 000 b. Even a small amount of gadolinium considerably decreases f (see (19.14)). Although ^{157}Gd is continuously destroyed ("burned") by n,γ-reactions which convert it to ^{158}Gd ($\sigma_{n,\gamma}$ 2.5 b) during operation of the reactor, at the same time fission product poisons are formed and fissile nuclei used-up resulting in a compensating effect, which maintains an acceptable value of k_∞. Another possibility to regulate k, mainly used in PWRs, is the addition of a suitable amount of a water soluble n-absorber to the moderator – coolant, *chemical shim control*. When burnup increases, the concentration of the soluble absorber is gradually reduced in order to compensate the poisoning from the fission products and the consumption of fissile atoms.

Because many elements contain isotopes with high neutron capture cross sections for thermal neutrons all materials in the reactor must be extremely pure in order to keep k_∞ as large as possible.

An important factor is the temperature dependence of k. A temperature increase usually has little effect on $\eta\varepsilon$, but f usually increases (i.e. becomes close to 1) because of decreasing moderator density and average neutron energy increase; this causes more neutrons to be captured in ^{239}Pu, which has a low energy (0.3 eV) fission resonance. On the other hand, p decreases because thermal vibrations of atoms cause a *doppler broadening* of the n,γ resonances in the epithermal region, mostly for ^{238}U. UO_2 fuel has a relatively poor heat conductivity causing the center temperature in the fuel pins to vary strongly with the reactor power level, cf. Ch. 21. This gives the doppler broadening of resonances

in the fuel a high fission rate dependence. The doppler broadening also increases capture in the control rods. These latter effects normally dominate over the increase in f, so that k_∞ decreases slightly with increasing reactor power and temperature. This is referred to as a *negative temperature coefficient*. In a BWR a negative temperature coefficient is also caused by the bubbles of steam (*void*) occurring in the moderator, which reduces the thermalization of neutrons while increasing neutron leakage rate. In BWRs the effect of the void normally dominates over doppler broadening in limiting the reactor power. As the reactor tends to keep the void fraction and fuel temperature constant, the power level of a BWR (and to a smaller degree also of a PWR, where mostly the average fuel temperature is affected) can be adjusted by varying the speed of the coolant circulation pumps at a constant control rod setting.

These are important safety features of a reactor. If k increases, so do the fission rate and temperature. If the temperature coefficient was positive, k would increase further, leading to further temperature increase, etc. However, with a negative temperature coefficient the reactor controls itself: an increase in power (and thus temperature) decreases k, which tends to limit the power increase and vice versa. The temperature coefficient is usually given in $\%\Delta k/k$ per °C.

19.7. NEUTRON LEAKAGE AND CRITICAL SIZE

All practical reactors have some leakage of neutrons out of the reactor core. This leakage Λ is approximately described by the fast and thermal leakage factors Λ_f and Λ_{th} respectively:

$$\Lambda = \Lambda_f \Lambda_{th} \tag{19.15}$$

The effect of this leakage is included in Figure 19.6, and results in the number of neutrons in the second generation being reduced, from the 112 calculated previously for k_∞, to $N_2 = 101$. Thus $k_{eff} = 1.01$ and $\Lambda = 0.9$, i.e. 10% of the neutrons are lost through leakage. The example in Figure 19.6 refers to a small air cooled and graphite moderated natural uranium research reactor. In nuclear power stations k_∞ is much larger due to the large η-values (because of the enriched fuel needed to sustain operation for long periods at high power between refuelings), while the leakage is much smaller, $\Lambda \geq 0.97$ (e.g. see Table 19.4).

The leakage is a function of the geometrical arrangement of reactor core and reflector and of the average distance the neutron travels after formation until it causes a new fission:

$$\Lambda_f = e^{-B^2\tau} \tag{19.16}$$

$$\Lambda_{th} = \left(1 + B^2 L^2\right)^{-1} \tag{19.17}$$